Assignment 6

Week 6

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Problem 1. Power of Quantum Register

At the time of writing, the whole world has less than 10^{21} byte of memory. If we need two bytes to store one coefficient of a quantum state (assume one for the real part and one for the imaginary part), what is the maximum size of the quantum register state the world's memory can store?

Solution

What has to be stored. A register of n qubits lives in $\mathbb{C}^{2^n}$, so a general state

$$|\psi\rangle = \sum_{k=0}^{2^n-1} c_k \, |k\rangle$$

is a list of 2^n complex coefficients. At two bytes each, the storage cost is

$$B(n) = 2 \cdot 2^n = 2^{\,n+1} \text{ bytes.}$$

Solving the constraint. We need $B(n) \le 10^{21}$, so

$$2^{\,n+1} \le 10^{21} \quad \Longleftrightarrow \quad n+1 \le \log_2 10^{21} = 21 \log_2 10 \approx 69.76 \quad \Longleftrightarrow \quad n \le 68.76.$$

Since n must be an integer, $n_{\max} = 68$.
Check.

$$B(68) = 2^{69} \approx 5.90 \times 10^{20} \le 10^{21}, \qquad B(69) = 2^{70} \approx 1.18 \times 10^{21} > 10^{21}.$$

The entire world's memory – just under one zettabyte – therefore stores the state of a register of at most **68 qubits** (and nothing else).

> **Remark.** The bound is brutal because $B(n)$ doubles with every extra qubit: going from 68 to 69 qubits would need a second Earth's worth of memory. Storing the *state vector* is what is expensive, not building the qubits – a 68-qubit chip is ordinary hardware, while writing down what it is doing is already beyond every computer on the planet. This gap is exactly why quantum computers are worth building.

The notebook (**Appendix 1**) confirms $n_{\max} = \left\lfloor \log_2 \left(10^{21}/2\right) \right\rfloor = 68$.

Problem 2. Bell States

Prove that all Bell states are entangled.

Proof

Setup. The four Bell states are

$$|\phi^{\pm}\rangle = \tfrac{1}{\sqrt{2}} \left(|00\rangle \pm |11\rangle\right), \qquad |\psi^{\pm}\rangle = \tfrac{1}{\sqrt{2}} \left(|01\rangle \pm |10\rangle\right).$$

A two-qubit state is *separable* (i.e. not entangled) exactly when it can be written as a single tensor product of one-qubit states; it is *entangled* otherwise. Expanding a general product,

$$(a\,|0\rangle + b\,|1\rangle) \otimes (c\,|0\rangle + d\,|1\rangle) = ac\,|00\rangle + ad\,|01\rangle + bc\,|10\rangle + bd\,|11\rangle,$$

so $|\chi\rangle = \sum_{i,j} c_{ij}\,|ij\rangle$ is separable if and only if $c_{00} = ac$, $c_{01} = ad$, $c_{10} = bc$, $c_{11} = bd$ for some $a, b, c, d \in \mathbb{C}$.
A criterion. Any such factorization forces

$$c_{00}c_{11} - c_{01}c_{10} = (ac)(bd) - (ad)(bc) = abcd - abcd = 0,$$

that is, the coefficient matrix $C = \begin{pmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{pmatrix}$ must be singular. Contrapositively, **if** $\det C \neq 0$ **the state is entangled.**
Applying it to the four Bell states.

State	c_{00}	c_{01}	c_{10}	c_{11}	$\det C = c_{00}c_{11} - c_{01}c_{10}$	entangled?
$\lvert\phi^+\rangle$	$\frac{1}{\sqrt{2}}$	0	0	$\frac{1}{\sqrt{2}}$	$+\frac{1}{2}$	yes
$\lvert\phi^-\rangle$	$\frac{1}{\sqrt{2}}$	0	0	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{2}$	yes
$\lvert\psi^+\rangle$	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{2}$	yes
$\lvert\psi^-\rangle$	0	$\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$	0	$+\frac{1}{2}$	yes

Every determinant is $\pm\frac{1}{2} \neq 0$, so no Bell state is a product state: all four are entangled. ■

The same argument written out for $|\phi^+\rangle$. Suppose, for contradiction, that

$$\tfrac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) = \left(a\,|0\rangle + b\,|1\rangle\right) \otimes \left(c\,|0\rangle + d\,|1\rangle\right).$$

Matching coefficients gives $ac = \frac{1}{\sqrt{2}}$, $ad = 0$, $bc = 0$, $bd = \frac{1}{\sqrt{2}}$. From $ac \neq 0$ we get $a \neq 0$ and $c \neq 0$; then $ad = 0$ forces $d = 0$ and $bc = 0$ forces $b = 0$, so $bd = 0 \neq \frac{1}{\sqrt{2}}$ – a contradiction. The other three states fail in exactly the same way (for $|\psi^\pm\rangle$ the roles of the vanishing and non-vanishing pairs are swapped).

> **Remark.** Entanglement entropy gives the same verdict quantitatively: tracing out one qubit of any Bell state leaves $\rho_A = \frac{1}{2}I$, the maximally mixed state, so $S = 1$ bit – the largest value a two-qubit state can reach (**Problem 6**). A product state would instead give $S = 0$ (**Problem 5**).

Problem 3. Profiting using Entanglement

There is a strange trillionaire living on Pluto. She invests based on the measurement of electron spin every month. If it is a spin up, she will invest in the gold on the Earth. If it is spin down, she will invest in silver on the Earth. An investor was able to entangle his electron with her electron to be measured next month secretly. Once the measurement is performed, the investor knows that his electron will collapse to the same spin states, and he will buy the corresponding commodity before anyone else who receives the result through a radio wave from Pluto. Is this really useful? In other words, is this a riskless investment for the investor on Earth?

Solution

Short answer. No. The correlation he is exploiting is real, but it is nothing more than a shared coin flip, and it is certainly not riskless.

What quantum mechanics does give him. Say the pair is prepared in $|\phi^+\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle\right)$. Whoever measures first gets $\uparrow$ or $\downarrow$ with probability $\frac{1}{2}$ each, and the partner electron then agrees with certainty. So yes – by measuring his own electron he learns her outcome without waiting for the radio wave, which takes hours to cross the solar system.
What it does not give him. Nothing was *transmitted*. His electron alone is described by the reduced density matrix

$$\rho_A = \mathrm{Tr}_B\left(|\phi^+\rangle\langle\phi^+|\right) = \tfrac{1}{2}I,$$

which is the same maximally mixed state whether or not she has already measured (**Problem 6**). He cannot tell from his side *when* – or even *whether* – the measurement on Pluto has happened, and she cannot steer her outcome to encode a choice. This is the no-signalling theorem: entanglement never carries information faster than light.
So what is left is pre-shared randomness. The trillionaire decides nothing; the electron does, at random. The investor's "advantage" is exactly the advantage he would have had by tossing a fair coin on Earth, sealing the result in an envelope, and shipping her a copy before she left – a purely classical arrangement with the same correlations. Quantum mechanics only outperforms shared randomness when the two parties measure along *different* axes (Bell-inequality experiments), which

is not what is happening here.

Why it is not riskless.

- **Knowing the trade is not knowing the return.** He learns *which* commodity she buys, not what it will be worth. Gold can fall even after a trillionaire buys it, and the market risk – the only risk an investor actually cares about – is untouched.
- **The measurement axes must agree.** Perfect $\uparrow\uparrow$ / $\downarrow\downarrow$ correlation holds only when both measure along the same direction. If she measures along $\hat{x}$ while he measures along $\hat{z}$, each of his outcomes is 50/50 and tells him nothing (**Assignment 4, Problem 3**). He must know her apparatus exactly.
- **Decoherence.** One half of an entangled pair has to stay coherent for a month while travelling to Pluto. Any stray interaction – a magnetic field, a collision, a measurement by someone else – entangles the electron with its environment and degrades the correlation into noise. Worse, nothing in his own statistics reveals that this has happened: he still sees 50/50, so he would bet on a broken channel with full confidence.
- **Order of measurement.** If he measures first, *his* outcome is the random one and hers merely agrees, so he still has to assume her electron is untouched until then. One earlier measurement by anybody else breaks the link silently.

Conclusion. The scheme does deliver a bit of information before the radio wave arrives, but that bit is a fair coin flip that a classical sealed envelope would have supplied just as well, and it says nothing about prices. Physically it is fragile, and financially it is not riskless at all.

Problem 4. Inner Product between Two Spaces

What is the inner product between the basis states of two electron spins? For example, what is the inner products of $|0\rangle_1$ and $|0\rangle_0$?

Solution

Short answer. There is none: $\langle 0_1|0_0\rangle$ is **not defined**, because the two kets live in different Hilbert spaces.
Why. An inner product is a map $\mathcal{H}\times\mathcal{H}\to\mathbb{C}$; both arguments must come from the *same* space. Electron 1's spin lives in $\mathcal{H}_1\cong\mathbb{C}^2$ and electron 0's in $\mathcal{H}_0\cong\mathbb{C}^2$. These are isomorphic copies of $\mathbb{C}^2$, but they are not the same space, and no physical process identifies a vector of one with a vector of the other. Writing $\langle 0_1|0_0\rangle = 1$ because "both look like $(1,0)^\top$" is a category error: it treats a label as though it were a coordinate.
What the symbol can mean instead. A bra of space 1 meeting a ket of space 0 does not contract – they act on different factors, so they simply commute and stand side by side,

$$\langle 0|_1\,|0\rangle_0 = |0\rangle_0\,\langle 0|_1\,,$$

and the right-hand side is a perfectly good *operator*: a linear map $\mathcal{H}_1\to\mathcal{H}_0$, not a number.
What can be computed. Two spaces are joined by the *tensor* product, $\mathcal{H}_1\otimes\mathcal{H}_0$, and the inner product there factorizes with each bra meeting the ket of its own space (**Assignment 5, Problem 9**):

$$\langle f_1\otimes g_0 \mid e_1\otimes k_0\rangle = \langle f|e\rangle_1\ \langle g|k\rangle_0\,.$$

Only then does a number come out, for instance

$$\langle 0_1\,0_0 \mid 0_1\,0_0\rangle = \langle 0|0\rangle_1\,\langle 0|0\rangle_0 = 1\cdot 1 = 1, \qquad \langle 0_1\,0_0 \mid 0_1\,1_0\rangle = \langle 0|0\rangle_1\,\langle 0|1\rangle_0 = 1\cdot 0 = 0.$$

Moral. States of different subsystems combine with $\otimes$, never with $\langle\cdot|\cdot\rangle$. In particular $|0\rangle_1$ and $|0\rangle_0$ are neither orthogonal nor parallel – those words have no meaning across two spaces. The four two-qubit basis states $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ *are* orthonormal, but that is a statement inside the single space $\mathcal{H}_1\otimes\mathcal{H}_0$.

Problem 5. Entanglement Entropy of a Product State

Find the density matrix of $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$. Then find its entanglement entropy.

Hint: It should be zero because it is a product state. Can you also factorize it into the product of two vectors in the $\mathbb{C}^2$ space?

Note that: its density matrix is not diagonal. Try to diagonalize it first before taking the logarithmic values.

Solution

The density matrix. With $|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \frac{1}{2}(1,1,1,1)^\top$,

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix} \cdot \frac{1}{2}\begin{pmatrix}1 & 1 & 1 & 1\end{pmatrix} = \frac{1}{4}\begin{pmatrix}1 & 1 & 1 & 1\\ 1 & 1 & 1 & 1\\ 1 & 1 & 1 & 1\\ 1 & 1 & 1 & 1\end{pmatrix}.$$

Factorization. Grouping by the first qubit,

$$|\psi\rangle = \tfrac{1}{2}[|0\rangle(|0\rangle + |1\rangle) + |1\rangle(|0\rangle + |1\rangle)] = \tfrac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) = |+\rangle \otimes |+\rangle,$$

with $|+\rangle = \frac{1}{\sqrt{2}}(1,1)^\top$. It is indeed a product state, so $\rho = (|+\rangle\langle+|) \otimes (|+\rangle\langle+|)$.

Reduced density matrix. Writing ρ in 2×2 blocks labelled by the A indices, with $J = \begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$,

$$\rho = \frac{1}{4}\begin{pmatrix}J & J\\ J & J\end{pmatrix}, \qquad (\rho_A)_{aa'} = \mathrm{Tr}\,(\mathrm{block}_{aa'}) = \tfrac{1}{4}\mathrm{Tr}\,J = \tfrac{1}{2},$$

so

$$\rho_A = \mathrm{Tr}_B(\rho) = \frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} = |+\rangle\langle+|,$$

exactly the first factor – as it must be for a product state. Note it is **not** diagonal.

Diagonalize before taking the logarithm. The characteristic equation of ρ_A is

$$\det(\rho_A - \lambda I) = \left(\tfrac{1}{2} - \lambda\right)^2 - \tfrac{1}{4} = \lambda^2 - \lambda = 0 \quad \Longrightarrow \quad \lambda_1 = 1,\ \lambda_2 = 0,$$

with eigenvectors $|+\rangle$ and $|-\rangle$. Hence $\rho_A = U\,\mathrm{diag}(1,0)\,U^\dagger$ with $U = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$, and the logarithm acts on the eigenvalues only.

Entanglement entropy.

$$S = -\mathrm{Tr}\,(\rho_A \log_2 \rho_A) = -\sum_i \lambda_i \log_2 \lambda_i = -(1\cdot\log_2 1) - (0\cdot\log_2 0) = 0 - 0 = 0,$$

using the standard convention $0\log_2 0 := \lim_{x\to 0^+} x\log_2 x = 0$.

The entropy vanishes, matching the hint: $\rho_A = |+\rangle\langle+|$ is a *pure* state, so qubit A is in a definite state of its own and knows nothing about qubit B. The notebook (**Appendix 1**) reproduces ρ, the factorization $|\psi\rangle = |+\rangle \otimes |+\rangle$, the eigenvalues $\{1,0\}$, and $S = 0$.

> **Remark.** This is why the diagonalization matters. Taking logarithms of the entries of ρ_A would mean evaluating $\log_2 \frac{1}{2}$ four times and give a nonzero (and meaningless) number. In the eigenbasis, $\log_2 \rho_A = \mathrm{diag}(0, -\infty)$, but the divergent entry is multiplied by the eigenvalue 0 and contributes nothing.

Problem 6. Partial Tracing

Find the entanglement entropy of $|\phi^-\rangle$ but trace the B space. You should get the same answer. The equation now becomes:

$$S_B = -\mathrm{Tr}_A\left(\rho_A \log_2 \rho_A\right), \text{ where } \rho_A = \mathrm{Tr}_B\left(\rho\right)$$

Solution

The density matrix. With $|\phi^-\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle - |11\rangle\right)$,

$$\rho = |\phi^-\rangle\langle\phi^-| = \tfrac{1}{2}\left(|00\rangle\langle 00| - |00\rangle\langle 11| - |11\rangle\langle 00| + |11\rangle\langle 11|\right) = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}.$$

Tracing out B. The partial trace over B contracts only the second slot,

$$\rho_A = \mathrm{Tr}_B\left(\rho\right) = \sum_{b\in\{0,1\}} \left(I \otimes \langle b|\right)\rho\left(I \otimes |b\rangle\right), \quad \text{so} \quad \mathrm{Tr}_B\left(|ab\rangle\langle a'b'|\right) = |a\rangle\langle a'|\,\langle b'|b\rangle = \delta_{bb'}\,|a\rangle\langle a'|\,.$$

The cross terms have $b \neq b'$ and die, e.g. $\mathrm{Tr}_B\left(|00\rangle\langle 11|\right) = \delta_{01}\,|0\rangle\langle 1| = 0$. Only the diagonal terms survive:

$$\rho_A = \tfrac{1}{2}\left(|0\rangle\langle 0| + |1\rangle\langle 1|\right) = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2}I.$$

Entanglement entropy. ρ_A is already diagonal, with eigenvalues $\lambda_1 = \lambda_2 = \frac{1}{2}$, so $\log_2 \rho_A = \mathrm{diag}\left(-1,-1\right)$ and

$$S_B = -\mathrm{Tr}_A\left(\rho_A \log_2 \rho_A\right) = -\sum_i \lambda_i \log_2 \lambda_i = -\left(\tfrac{1}{2}\log_2\tfrac{1}{2} + \tfrac{1}{2}\log_2\tfrac{1}{2}\right) = -2\cdot\tfrac{1}{2}\cdot(-1) = 1 \text{ bit}.$$

Same answer as tracing A. Tracing the other space gives $\rho_B = \mathrm{Tr}_A\left(\rho\right) = \frac{1}{2}I$ by the identical calculation, so $S_A = S_B = 1$. This is no coincidence: for a *pure* bipartite state the Schmidt decomposition $|\chi\rangle = \sum_k \sqrt{p_k}\,|u_k\rangle_A \otimes |v_k\rangle_B$ makes ρ_A and ρ_B share the same nonzero spectrum $\{p_k\}$, hence the same entropy. Entanglement is a property of the *pair*; it cannot matter which half you look through.

Getting $\rho_A = \frac{1}{2}I$, the maximally mixed state, means qubit A on its own is completely random – $S = 1$ bit is the largest value a single qubit can carry, so $|\phi^-\rangle$ is maximally entangled (compare $S = 0$ for the product state of **Problem 5**). The notebook (**Appendix 1**) confirms $\rho_A = \frac{1}{2}I$, eigenvalues $\{\frac{1}{2},\frac{1}{2}\}$, and $S_A = S_B = 1$.

Problem 7. Unitarity of the NOT Gate

Use hand calculation and colab to show that a 2-qubit NOT gate is unitary.

Solution

The gate. A NOT gate on a 2-qubit register flips both qubits, so it is $X \otimes X$ with $X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$:

$$X \otimes X = \begin{pmatrix} 0 \cdot X & 1 \cdot X \\ 1 \cdot X & 0 \cdot X \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

which maps $|00\rangle \leftrightarrow |11\rangle$ and $|01\rangle \leftrightarrow |10\rangle$, as a double NOT should.

Hand calculation. The matrix is real and symmetric, so $(X \otimes X)^\dagger = X \otimes X$ and unitarity reduces to squaring it:

$$(X \otimes X)^\dagger (X \otimes X) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = I_4,$$

since row i of the first factor picks out row $5-i$ of the second, which has its single 1 back in column i. The same holds for $(X \otimes X)(X \otimes X)^\dagger$, so $X \otimes X$ is unitary. ■

Without writing any matrix. The mixed-product property $(A \otimes B)(C \otimes D) = AC \otimes BD$ and $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$ give it in one line:

$$(X \otimes X)^\dagger (X \otimes X) = \left(X^\dagger \otimes X^\dagger\right)(X \otimes X) = X^\dagger X \otimes X^\dagger X = I \otimes I = I_4,$$

using $X^\dagger X = X^2 = I$ (**Assignment 3**). A tensor product of unitaries is always unitary, which is why single-qubit gates can be applied in parallel.

Colab. The notebook (**Appendix 1**) builds $X \otimes X$ with `np.kron` and checks both products against I_4:

```
X = np.array([[0, 1], [1, 0]])
NOT2 = np.kron(X, X)

print(np.allclose(NOT2.conj().T @ NOT2, np.eye(4)),
      np.allclose(NOT2 @ NOT2.conj().T, np.eye(4)))
```

which prints `True True`, confirming the hand calculation.

> **Remark.** $X \otimes X$ is both Hermitian and unitary, so it is its own inverse: applying NOT twice returns the register to its original state, exactly as classical NOT does.

Problem 8. Unitarity of the CNOT Gate

Use hand calculation and colab to show that the CNOT gate is unitary.

Solution

The gate. CNOT flips the target qubit when the control qubit is 1 and does nothing otherwise. Taking the first (most significant) qubit as the control,

$$|00\rangle \mapsto |00\rangle, \quad |01\rangle \mapsto |01\rangle, \quad |10\rangle \mapsto |11\rangle, \quad |11\rangle \mapsto |10\rangle, \qquad \mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Hand calculation. The matrix is real and symmetric, so $\mathrm{CNOT}^\dagger = \mathrm{CNOT}^\top = \mathrm{CNOT}$, and

$$\mathrm{CNOT}^\dagger \mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = I_4 :$$

the upper block is I_2 squared, and the lower block is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = I_2$. The same product in the other order gives I_4 as well, so CNOT is unitary. ■

In block form. Writing the gate as "identity on the target if the control is $|0\rangle$, X on the target if it is $|1\rangle$",

$$\mathrm{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X,$$

and since $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ are orthogonal projectors ($\langle 0|1\rangle = 0$), the cross terms vanish:

$$\mathrm{CNOT}^\dagger \mathrm{CNOT} = |0\rangle\langle 0| \otimes I^\dagger I + |1\rangle\langle 1| \otimes X^\dagger X = (|0\rangle\langle 0| + |1\rangle\langle 1|) \otimes I = I \otimes I = I_4.$$

Why this was inevitable. CNOT merely permutes the four basis states $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, so its columns are the standard basis vectors in the order $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_4, \mathbf{e}_3$ – still orthonormal. A matrix that maps an orthonormal basis to an orthonormal basis is unitary (**Assignment 4, Problem 8**). Every reversible classical gate becomes a unitary quantum gate this way.

Colab. The notebook (**Appendix 1**) checks both products:

```
CNOT = np.array([[1, 0, 0, 0],
                 [0, 1, 0, 0],
                 [0, 0, 0, 1],
                 [0, 0, 1, 0]])

print(np.allclose(CNOT.conj().T @ CNOT, np.eye(4)),
      np.allclose(CNOT @ CNOT.conj().T, np.eye(4)))
```

which prints `True True`, confirming the hand calculation.

Problem 9. Implementation of a Quantum Circuit

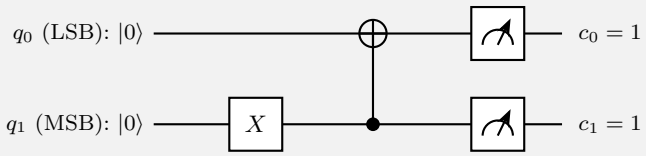
Use IBM-Q, by choosing "IBM Quantum Composer" to implement a 2-qubit circuit in which a 1-qubit NOT gate is applied to the MSB and then a CNOT gate is applied to the two qubits with the MSB as the control qubit. Refer to Assignment 1, Problem 3 for how to construct a circuit. Firstly, based on what you have learned, what result do you expect? Add measurement to check if you get the desired result.

Hint: since the input begins with $|00\rangle$, the output is 100% $|11\rangle$.

Note that: in IBM-Q, the bottom qubit is the MSB.

Solution

The circuit. In the Composer the bottom wire is the MSB, so the X gate and the control dot both go on q_1, and the CNOT target is q_0:



Expected result (before running). Write the register as $|q_1 q_0\rangle$, MSB first, starting from $|00\rangle$.

$$\begin{aligned} \text{NOT on the MSB:} \quad & (X \otimes I)\,|00\rangle = (X\,|0\rangle) \otimes (I\,|0\rangle) = |1\rangle \otimes |0\rangle = |10\rangle\,, \\ \text{then CNOT (control = 1):} \quad & \text{CNOT}\,|10\rangle = |1\rangle \otimes (X\,|0\rangle) = |1\rangle \otimes |1\rangle = |11\rangle\,. \end{aligned}$$

In column form, with the matrices of **Problems 7 and 8**,

$$\begin{aligned} \text{CNOT}\,(X \otimes I)\,|00\rangle &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |11\rangle\,. \end{aligned}$$

The output is the basis state $|11\rangle$, not a superposition, so the measurement is deterministic:

$$\text{P}\,(|11\rangle) = |\langle 11|\psi\rangle|^2 = 1 = 100\%, \qquad \text{P}\,(|00\rangle) = \text{P}\,(|01\rangle) = \text{P}\,(|10\rangle) = 0.$$

Adding the two measurement operations and running the job returns the bit string `11` in every shot, matching the hint. The notebook (**Appendix 1**) reproduces the same circuit numerically and prints $\text{P}(|11\rangle) = 100.00\%$.

> **Remark.** This is the classical logic "set the control, then copy it onto the target" done reversibly – no superposition is ever created, which is why the histogram has a single bar. Had a Hadamard replaced the X, the same CNOT would have produced the Bell state $|\phi^+\rangle$ of **Problem 2** and a 50/50 histogram over `00` and `11`. On real hardware a few percent of the shots still land on the other three strings, from gate and readout errors rather than from the physics.

APPENDIX 1

Assignment 06 | Code Solution

assignment-06-code

September 16, 2026

1 Assignment 06

1.1 Import External Libraries

[1]:
```
import numpy as np
import sympy as sp

from IPython.display import display, Math
```

1.2 1. Power of Quantum Register

[2]:
```
# --- Problem 1: how many qubits fit in the world's memory? ---
WORLD_BYTES = 1e21    # the whole world has less than 10^21 bytes of memory
BYTES_PER_COEFF = 2   # one byte for the real part, one for the imaginary part

# an n-qubit register has 2^n coefficients  ->  2 * 2^n = 2^(n+1) bytes
n_max = int(np.floor(np.log2(WORLD_BYTES / BYTES_PER_COEFF)))

display(Math(r"n_{\max} = \left\lfloor \log_2 \frac{10^{21}}{2} \right\rfloor = "
             + str(n_max)))

for n in (n_max, n_max + 1):
    needed = BYTES_PER_COEFF * 2.0 ** n
    verdict = "fits" if needed <= WORLD_BYTES else "too big"
    print(f"n = {n} qubits  ->  {needed:.3e} bytes  ({verdict})")
```

$$n_{\max} = \left\lfloor \log_2 \frac{10^{21}}{2} \right\rfloor = 68$$

```
n = 68 qubits  ->  5.903e+20 bytes  (fits)
n = 69 qubits  ->  1.181e+21 bytes  (too big)
```

1.3 5. Entanglement Entropy of a Product State

[3]:
```
# --- Problem 5: density matrix and entanglement entropy of a product state ---
ket0, ket1 = np.array([1, 0]), np.array([0, 1])


def partial_trace_B(rho):
    """Tr_B(rho) for a 2-qubit rho: contract the B index (rows/cols are (a,b))."""
    return np.einsum("abcb->ac", rho.reshape(2, 2, 2, 2))


def entanglement_entropy(rho_A, tol=1e-12):
    """S = -Tr(rho log2 rho), taken on the eigenvalues, with 0 log 0 := 0."""
    eigvals = np.linalg.eigvalsh(rho_A).real
```

```
    eigvals = eigvals[eigvals > tol]
    return float(-np.sum(eigvals * np.log2(eigvals))) + 0.0


psi = np.ones(4) / 2   # (|00> + |01> + |10> + |11>) / 2
rho = np.outer(psi, psi.conj())
display(Math(r"\rho = \ket{\psi}\bra{\psi} = " + sp.latex(sp.Matrix(np.round(rho, 4)))))

# it is a product state:  |psi> = |+> (x) |+>
plus = np.array([1, 1]) / np.sqrt(2)
print("Problem 5  --  |psi> == |+> (x) |+> ?", np.allclose(psi, np.kron(plus, plus)))

print("---------------")

rho_A = partial_trace_B(rho)
display(Math(r"\rho_A = \mathrm{Tr}_B(\rho) = " + sp.latex(sp.Matrix(np.round(rho_A, 4)))))
print("Problem 5  --  eigenvalues of rho_A:", np.round(np.linalg.eigvalsh(rho_A).real, 4))
print("Problem 5  --  S = -Tr(rho_A log2 rho_A) =", entanglement_entropy(rho_A))
```

$$\rho = |\psi\rangle\langle\psi| = \begin{bmatrix} 0.25 & 0.25 & 0.25 & 0.25 \\ 0.25 & 0.25 & 0.25 & 0.25 \\ 0.25 & 0.25 & 0.25 & 0.25 \\ 0.25 & 0.25 & 0.25 & 0.25 \end{bmatrix}$$

```
Problem 5  --  |psi> == |+> (x) |+> ? True
---------------
```

$$\rho_A = \mathrm{Tr}_B(\rho) = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

```
Problem 5  --  eigenvalues of rho_A: [0. 1.]
Problem 5  --  S = -Tr(rho_A log2 rho_A) = 0.0
```

1.4 6. Partial Tracing

[4]:
```
# --- Problem 6: entanglement entropy of |phi-> by tracing out the B space ---
phi_minus = (np.kron(ket0, ket0) - np.kron(ket1, ket1)) / np.sqrt(2)
rho = np.outer(phi_minus, phi_minus.conj())
display(Math(r"\rho = \ket{\phi^-}\bra{\phi^-} = " + sp.latex(sp.Matrix(np.round(rho, 4)))))

rho_A = partial_trace_B(rho)
display(Math(r"\rho_A = \mathrm{Tr}_B(\rho) = " + sp.latex(sp.Matrix(np.round(rho_A, 4)))))
print("Problem 6  --  eigenvalues of rho_A:", np.round(np.linalg.eigvalsh(rho_A).real, 4))
print("Problem 6  --  S_B = -Tr_A(rho_A log2 rho_A) =", entanglement_entropy(rho_A))

print("---------------")

# tracing the other space instead must give the same number (Schmidt decomposition)
rho_B = np.einsum("abad->bd", rho.reshape(2, 2, 2, 2))
print("Problem 6  --  S_A = -Tr_B(rho_B log2 rho_B) =", entanglement_entropy(rho_B))
```

$$\rho = |\phi^-\rangle\langle\phi^-| = \begin{bmatrix} 0.5 & 0.0 & 0.0 & -0.5 \\ 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 \\ -0.5 & 0.0 & 0.0 & 0.5 \end{bmatrix}$$

$$\rho_A = \mathrm{Tr}_B(\rho) = \begin{bmatrix} 0.5 & 0.0 \\ 0.0 & 0.5 \end{bmatrix}$$

```
Problem 6  --  eigenvalues of rho_A: [0.5 0.5]
Problem 6  --  S_B = -Tr_A(rho_A log2 rho_A) = 1.0
```

```
---------------
Problem 6  --  S_A = -Tr_B(rho_B log2 rho_B) = 1.0
```

1.5 7. Unitarity of the NOT Gate

[5]:
```
# --- Problem 7: the 2-qubit NOT gate X (x) X is unitary ---
X = np.array([[0, 1], [1, 0]])
NOT2 = np.kron(X, X)

display(Math(r"X \otimes X = " + sp.latex(sp.Matrix(NOT2))))
display(Math(r"(X \otimes X)^\dagger (X \otimes X) = "
             + sp.latex(sp.Matrix(NOT2.conj().T @ NOT2))))

print("Problem 7  --  unitary?",
      np.allclose(NOT2.conj().T @ NOT2, np.eye(4)),
      np.allclose(NOT2 @ NOT2.conj().T, np.eye(4)))
```

$$X \otimes X = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$(X \otimes X)^\dagger (X \otimes X) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

```
Problem 7  --  unitary? True True
```

1.6 8. Unitarity of the CNOT Gate

[6]:
```
# --- Problem 8: the CNOT gate is unitary ---
CNOT = np.array([[1, 0, 0, 0],
                 [0, 1, 0, 0],
                 [0, 0, 0, 1],
                 [0, 0, 1, 0]])

display(Math(r"\mathrm{CNOT} = " + sp.latex(sp.Matrix(CNOT))))
display(Math(r"\mathrm{CNOT}^\dagger\,\mathrm{CNOT} = "
             + sp.latex(sp.Matrix(CNOT.conj().T @ CNOT))))

print("Problem 8  --  unitary?",
      np.allclose(CNOT.conj().T @ CNOT, np.eye(4)),
      np.allclose(CNOT @ CNOT.conj().T, np.eye(4)))
```

$$\mathrm{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\mathrm{CNOT}^\dagger\,\mathrm{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

```
Problem 8  --  unitary? True True
```

1.7 9. Implementation of a Quantum Circuit

[7]:
```
# --- Problem 9: NOT on the MSB, then CNOT with the MSB as the control ---
I2 = np.eye(2)
ket00 = np.kron(ket0, ket0)   # the register starts in |00>

state = CNOT @ (np.kron(X, I2) @ ket00)   # X on the MSB (first factor), then CNOT
display(Math(r"\mathrm{CNOT}\,(X \otimes I)\ket{00} = "
             + sp.latex(sp.Matrix(np.round(state, 4)).T)))

for label, amplitude in zip(["00", "01", "10", "11"], state):
    print(f"P(|{label}>) = {abs(amplitude) ** 2:.2%}")
```

$$\mathrm{CNOT}\,(X \otimes I)\,|00\rangle = \begin{bmatrix}0.0 & 0.0 & 0.0 & 1.0\end{bmatrix}$$

```
P(|00>) = 0.00%
P(|01>) = 0.00%
P(|10>) = 0.00%
P(|11>) = 100.00%
```
